

Yashpal Singh Chouhan

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ENGINEERING HIGHLIGHTS

- Architected and deployed 3 full-stack AI/LLM applications (RAG medical assistant, DSA AI coach, news bias detector) using FastAPI, Next.js, LangChain, ChromaDB, FAISS, and Google Gemini — all live in production
- Cut hallucinations in a medical QA system by grounding all responses in cited, page-level source chunks — 173 pages ingested, 532 vectors indexed via ChromaDB and HuggingFace embeddings

EDUCATION

VIT Bhopal University

Integrated M.Tech — Artificial Intelligence

— CGPA: 8.10 / 10.0

— Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, NLP, DBMS, OOP

Bhopal, India

2022 – 2027

EXPERIENCE

Data Analyst Intern

January 2026 – April 2026

CortexGiga (Remote)

- Processed and standardized 3 large structured datasets (10K+ rows each) in Pandas and NumPy — resolving null values, type inconsistencies, and duplicates — achieving 35% improvement in downstream pipeline accuracy
- Engineered 4 automated KPI dashboards in Python (Matplotlib, Seaborn) replacing manual Excel workflows; cut stakeholder turnaround from 2 days to under 3 hours — a 50% efficiency gain
- Established 15+ data quality validation standards with product and engineering teams, ensuring consistent integrity across the full analytics pipeline

PROJECTS

Medical RAG Assistant | *Python, FastAPI, LangChain, ChromaDB, Gemini API, HuggingFace, PyMuPDF*

- Designed a Retrieval-Augmented Generation system for medical document QA — parsed and indexed a 173-page clinical handbook into 532 semantic vectors using PyMuPDF, LangChain, and ChromaDB, with HuggingFace sentence-transformer embeddings for context-aware retrieval
- Exposed a FastAPI REST interface for document ingestion, semantic search, and Gemini-powered answer generation with page-level source citations, significantly reducing hallucination risk

DSA AI Coach | *Next.js, React, TypeScript, Tailwind CSS, FastAPI, Python, Gemini API, Vercel, Render* [GitHub](#) | [Live](#)

- Built and shipped a full-stack AI learning platform: Next.js frontend (Vercel) backed by a FastAPI service (Render); Gemini API generates personalized profiles, adaptive roadmaps, and lesson content from learner behavior
- Implemented Socratic AI mentoring — the system detects a user's solution approach and delivers targeted hints over direct answers, reinforcing conceptual understanding through guided problem-solving

Indian News Bias Detector | *Python, RAG, FAISS, Groq, LLaMA 3.3 70B, spaCy, Streamlit*

[GitHub](#)

- Scraped 300+ articles from 4 major Indian outlets across 15 political topics; applied a multi-layer NLP framing pipeline combining VADER sentiment, TextBlob subjectivity, and spaCy NER for source-level bias profiling
- Built a RAG pipeline pairing a FAISS vector store (SentenceTransformer embeddings) with Groq-hosted LLaMA 3.3 70B for context-grounded bias explanations; shipped an interactive Streamlit dashboard with bias heatmaps and topic-wise outlet comparison

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, TypeScript, SQL

Frontend: React.js, Next.js, Tailwind CSS, HTML, CSS

Backend & APIs: FastAPI, REST APIs, Git, GitHub

AI / ML / LLM: LangChain, RAG Pipelines, Google Gemini API, Groq API, Prompt Engineering, HuggingFace Embeddings, Sentence Transformers, Vector Databases (ChromaDB, FAISS), scikit-learn, TensorFlow, PyTorch

NLP: Sentiment Analysis, NER, Text Classification, spaCy, NLTK, VADER, TextBlob

Data & Analysis: Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, EDA

Databases & Tools: ChromaDB, MySQL, SQLite, Vercel, Render, Streamlit, VS Code, Jupyter Notebook

CERTIFICATIONS & ACHIEVEMENTS

- Applied Machine Learning in Python — University of Michigan (Coursera)
- Computer Vision — VIT Bhopal University (VITYarthi)
- Solved 100+ DSA problems on LeetCode/Codeforces — data structures, algorithmic optimization, graph algorithms
- **Research Intern, Kanchan Children Welfare & Educational Society:** Field assessments across 5+ rural schools; findings directly informed NGO welfare program planning and resource allocation